

# Rajalakshmi Engineering College

Name: GULZHAR AHAMAD  
Email: 240801090@rajalakshmi.edu.in  
Roll no: 240801090  
Phone: 9043074883  
Branch: REC  
Department: I ECE FA  
Batch: 2028  
Degree: B.E - ECE

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## NeoColab\_REC\_CS23231\_DATA STRUCTURES

### REC\_DS using C\_Week 1\_COD\_Question 5

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

Imagine you are tasked with developing a simple GPA management system using a singly linked list. The system allows users to input student GPA values, insertion should happen at the front of the linked list, delete record by position, and display the updated list of student GPAs.

##### ***Input Format***

The first line of input contains an integer  $n$ , representing the number of students.

The next  $n$  lines contain a single floating-point value representing the GPA of each student.

The last line contains an integer position, indicating the position at which a student record should be deleted. Position starts from 1.

### **Output Format**

After deleting the data in the given position, display the output in the format "GPA: " followed by the GPA value, rounded off to one decimal place.

Refer to the sample output for formatting specifications.

### **Sample Test Case**

Input: 4

3.8

3.2

3.5

4.1

2

Output: GPA: 4.1

GPA: 3.2

GPA: 3.8

### **Answer**

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
typedef struct Node{  
    float gpa;  
    struct Node* next;  
}Node;
```

```
void insert_front(Node** head,float gpa){  
    Node* new_node=(Node*)malloc(sizeof(Node));  
    new_node->gpa=gpa;  
    new_node->next=*head;  
    *head=new_node;  
}
```

```
void delete_by_position(Node** head,int position){  
    if(*head==NULL)return;  
    Node* temp=*head;  
    if(position==1){  
        *head=temp->next;
```

```

        free(temp);
        return;
    }
    Node* prev = NULL;
    for(int i=1;i<position;i++){
        prev=temp;
        if(temp->next==NULL)return;
        temp=temp->next;
    }
    prev->next=temp->next;
    free(temp);
}

void display(Node* head){
    Node* temp=head;
    while(temp!=NULL){
        printf("GPA: %.1f\n",temp->gpa);
        temp=temp->next;
    }
}

int main(){
    int n,pos;
    float gpa;
    Node* head=NULL;
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%f",&gpa);
        insert_front(&head,gpa);
    }
    scanf("%d",&pos);
    delete_by_position(&head,pos);
    display(head);
    return 0;
}

```

**Status : Correct**

**Marks : 10/10**